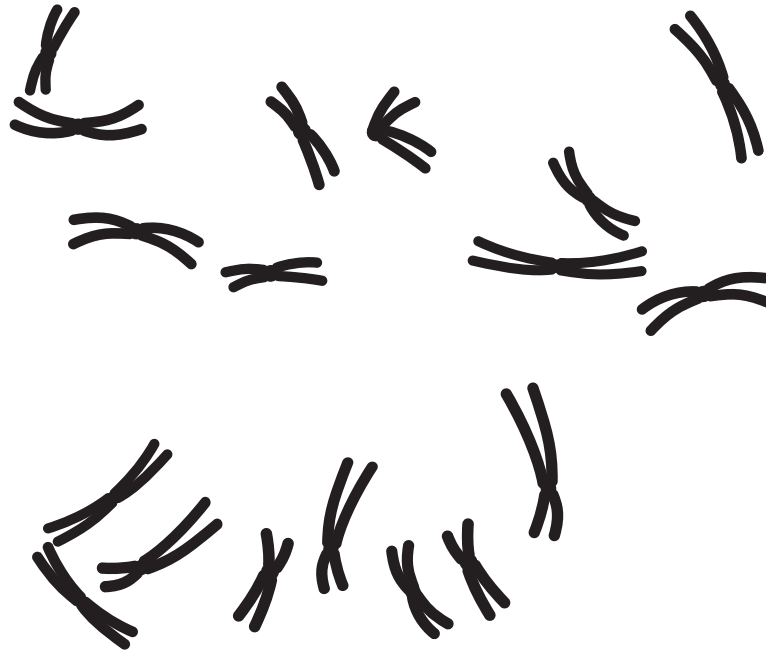


Answer all questions.

1. (a) The diagram below shows the chromosomes from a single cell of a Chinese daisy, *Aster shennongjiaensis*, during prophase of mitosis.



- (i) State the **haploid** number of chromosomes in this species. [1]

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- (ii) State the evidence from the diagram that the cell is undergoing mitosis and not meiosis. [1]

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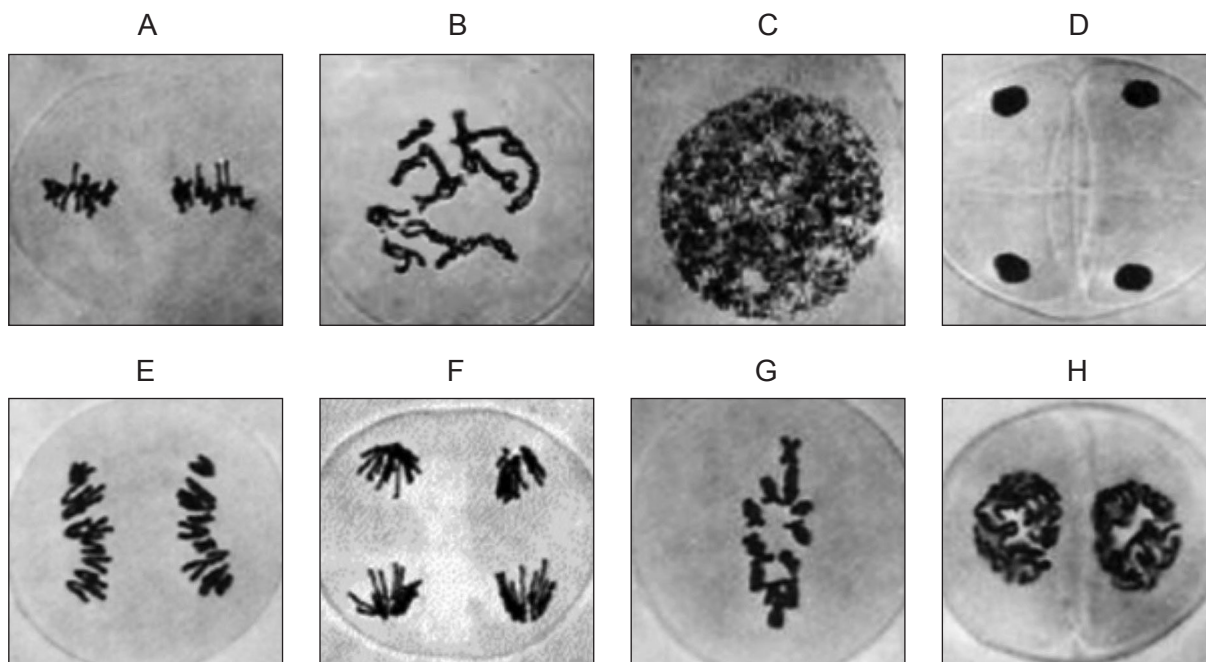
- (iii) When the tissues from the plant were removed they were stained before being viewed under the microscope. Explain how this makes the chromosomes visible. [1]

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(b) The photomicrographs below show different stages of meiosis in a different plant species.



- (i) State which organ of the plant could have been used to obtain the photomicrographs above. [1]

- (ii) Place the photomicrographs in the correct order. Some letters have been done for you. [1]

C					A	F	D
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- (iii) Use your knowledge of cell division to describe **two** differences between the stages labelled **E** and **F**. [2]

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- (iv) With reference to photomicrographs **B** and **G**, state how meiosis contributes to genetic variation in the species and explain why this is important. [3]

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